Random Variables: Expectation

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MATH 241

Week 7, Lecture #1 — Ch. 4.1-4.5

Reminders!

- Checkpoint #4 on **Thursday at beginning of class**
  - 15 minute quiz, open-note, on expectations and discrete random variables
  - Bring calculators!
- Homework #5 due on **Friday at 11:59pm**
  - Textbook and practice problems
  - Optional mid-semester survey
- Office hours after class today and on Friday from 10-11:30am

Today's Material

- Expectation, $E(X)$
  - What is it?
  - Explore its application to various random variables through calculations
- Properties of expectations
  - Linearity
  - Functions

Expectation

The **expected value** of a discrete random variable X whose PMF is $p_X(x)$, is defined as:

$$E(X) = \sum_{x:p_X(x)>0} x\, p_X(x)$$

- e.g. expectation, average, mean
- For each value of the support of X: take the product of the value x and its probability $P(X = x)$
- Can be referred to as μ_X

Expectation: Other Interpretations

$$E(X) = \sum_{x:p_X(x)>0} x\, p_X(x)$$

- Think of it as a weighted average
  - Some weight of $p_X(x)$ for each x
- Continuous version: replace summation with integration of all support values up to x
- Can substitute X for a function of a random variable
  - Will affect the expression x, not $p_X(x)$

Expectation: General Definition

For $s \in \mathcal{S}$, the sample space of the experiment, we can define the expectation of X as:

$$E(X) = \sum_{s \in \mathcal{S}} X(s)\, P(\{s\})$$

- Sum is over all possible outcomes s in sample space
- This provides a more in-depth interpretation of the expectation, but we will use the first definition for calculations

Example: Dice Rolls

Suppose X is the number we roll for a single die. Find the expectation.

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1. Find/compile support of X and $p_X(x)$:

$X = x$	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

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2. Piece together product of x and $p_X(x)$ for each value of support

$$\begin{aligned} E(X) &= \sum_x x\, p_X(x) \\ &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= 3.5 \end{aligned}$$

Example: Binomial Distribution

Say we flip four weighted coins independently and are curious about X, the number of heads I get in these four flips. The chance of getting heads is 60%. Find the expectation of X.

(1) Find/compile support of X and $p_X(x)$:

$X = x$	0	1	2	3	4
$P(X = x)$	0.4^4	$\binom{4}{1}0.6^1 0.4^3$	$\binom{4}{2}0.6^2 0.4^2$	$\binom{4}{3}0.6^3 0.4^1$	0.6^4
	$=0.0256$	$=0.1536$	$=0.3456$	$=0.3456$	$=0.1296$

Example: Binomial Distribution

Say we flip four weighted coins independently and are curious about X, the number of heads I get in these four flips. The chance of getting heads is 60%. Find the expectation of X.

(2) Piece together product of x and $p_X(x)$ for each value of support:

$$\begin{aligned} E(X) &= \sum_x x\, p_X(x) \\ &= 0(0.0256) + 1(0.1536) + 2(0.3456) + 3(0.3456) + 4(0.1296) \\ &= 2.4 \end{aligned}$$

Activity: Finding an Expectation

Let X be the number of club members who attend a monthly meeting, where the club has 6 members. Suppose we have the following probability distribution:

- $P(X = 0) = 0.02$
- $P(X = 1) = 0.07$
- $P(X = 2) = 0.11$
- $P(X = 3) = 0.2$
- $P(X = 4) = 0.23$
- $P(X = 5) = 0.29$
- $P(X = 6) = 0.08$

Find the expectation of X.

Properties: Same Distributions

If X and Y are from the same distribution:

$$E(X) = E(Y)$$

Properties: Linearity (Scaling and Shifting)

For any two random variables X and Y, we have the following properties for a linear combination of expectations:

$$E(X+Y) = E(X) + E(Y)$$
$$E(cX) = cE(X)$$

Activity: Expectations of Two RVs

Suppose you roll two dice, one back-to-back in the hopes of winning some money. On the first roll, you get 2 times the face value of the first roll X and on the second roll, you get 3 times the face value of the second roll Y. Find the expectation of your winnings.

Aside: Indicator Random Variables

An indicator random variable I_A for event A equals:

- 1 if A occurs
- 0 otherwise

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- 0 otherwise

$$E(A_I) = 1P(A) + 0P(A^C) = P(A)$$

- The probability of an event A equals the expected value of its indicator random variable, i.e. $P(A) = E(I_A)$.

Expectation of a Function of a Random Variable (LOTUS)

Suppose X is a discrete random variable and g is a function from $\mathbb{R} \to \mathbb{R}$, then:

$$E[g(X)] = \sum_x g(x) p_X(x),$$

where the sum is taken over the support of X, our original RV.

- Why isn't $g(x)$ included in our PMF?
  - Law of the Unconscious Statistician

Proof: LOTUS

We can prove the result as shown below using the more formal definition for expected value:

$$
\begin{aligned}
E[g(X)] &= \sum_{s \in \mathcal{S}} g(X(s)) P(\{s\}) \\
&= \sum_{x} \sum_{s : X(s) = x} g(X(s)) P(\{s\}) \\
&= \sum_{x} g(x) \sum_{s : X(s) = x} P(\{s\}) \\
&= \sum_{x} g(x) P(X = x) \\
&= \sum_{x} g(x) p_X(x)
\end{aligned}
$$

Proof for Linearity

Say we have random variables X and we want to find what $E(aX + b)$ is. Then,

$$\begin{aligned} E[aX + b] &= \sum_x (ax + b)p_X(x) \\ &= \sum_x axp_X(x) + \sum_x bp_X(x) \\ &= a\sum_x xp_X(x) + b\sum_x p_X(x) \\ &= aE(X) + b \end{aligned}$$

Proof for Linearity

This can also be shown for two certain functions:

$$E[X^n] = \sum_x x^n p_X(x)$$

$$E[(X-\mu)^2] = \sum_x (x-\mu)^2 p_X(x)$$

- First formula: definition for the nth moment of X
- Second formula: definition for the variance of X (i.e. $Var(X)$)

Example: Binomial Distribution, Revisited

Say we flip four weighted coins independently and are curious about X, the number of heads I get in these four flips. The chance of getting heads is 60%. Find the expectation of X.

For a common distribution: let's simplify the expectation formula for any binomial random variable:

$$\begin{aligned} E(X) &= E\left(\sum_{i=1}^{n} Y_i\right) \text{ where } Y_i \sim Bern(p) \\ &= \sum_{i=1}^{n} E(Y_i) \\ &= \sum_{i=1}^{n} (0 \times (1-p) + 1 \times p) \\ &= \sum_{i=1}^{n} p \\ &= np \end{aligned}$$

Activity: Geometric Distribution

Suppose we have some X that is geometrically distributed. Do the following:

1. Find the expression for the expectation of X.
   - Hint: Taylor series may be used here.
2. Say we have X_1, X_2, and X_3 that are geometrically distributed and have $p = 0.2, 0.5, 0.8$ respectively.
   - Find the expectation of each of these three random variables.
   - Discuss the relationship between p and the expectation for geometric random variables.

Next Time: Variance

$$Var(X) = E(X^2) - E^2(X)$$